

🔔 Israel's doctors perform first-ever AR, 3D eye socket surgery



Surgeons are seen examining a model skull ahead of eye surgery using AR technology (Credit: www.jpost.com)

Doctors at the Galilee Medical Center in Nahariya, Israel, have performed an eye socket fracture repair using augmented reality (AR) and 3D technologies. The surgery was on a 31-year-old patient who needed the floor of his eye socket repaired due to a serious fracture, which had led to double vision as well as impairment of the look and symmetry of the eyes.

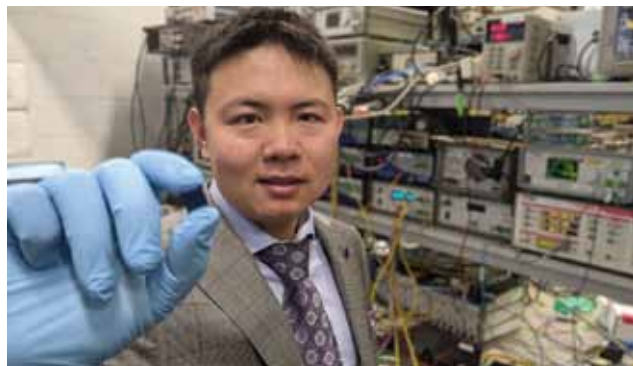
Using a software, a 3D-printed model of the patient's skull was built and, with the help of computerised tomography (CT) scan, a plate was designed that matched the patient's eye socket floor. This plate was then printed in titanium and placed beneath the eye using AR.

One of the surgeons wore Microsoft's HoloLens AR glasses, which connected to the computer software and thus both the patient's skull and plate models. The model was virtually placed over the patient's head and the surgeon then placed the plate in the correct position in real time. The surgery was successful and only took an hour and a half.

🔔 World's fastest optical neuromorphic processor developed for AI

An international team of researchers led by Swinburne University of Technology has demonstrated the world's fastest and most powerful optical neuromorphic processor for artificial intelligence (AI), which operates faster than ten trillion operations per second (TeraOPs/s) and is capable of processing ultra-large-scale data.

While state-of-the-art electronic processors such as the Google TPU can operate beyond 100 TeraOPs/s, this is done with tens of thousands of parallel processors. In contrast, the optical system demonstrated by the team uses a single processor and uses a new technique of simultaneously interleaving the data in time, wavelength, and spatial dimensions through an integrated micro-comb source. The breakthrough represents an enormous leap forward for neural networks and neuromorphic processing in general.



Dr Xingyuan (Mike) Xu with the integrated optical microcomb chip, which forms the core part of the optical neuromorphic processor (Credit: www.swinburne.edu.au)

🔔 Hydrogen fuel cells maximise drone flight time for humanitarian missions



DMI drones delivered emergency medical supplies to the Virgin Islands. A hydrogen fuel cell power pack made this possible by enabling over two hours of UAV flight time; four times longer than most battery-powered drones (Credit: www.vicorpower.com/press-room)

An average drone generally does about one hour of flight before it needs to be recharged, which can cause frequent interruptions during disaster relief missions. Now, South Korea based Doosan Mobility Innovation (DMI) has announced that they have successfully extended the flight time of drones to over two hours. The feat was achieved thanks to the incorporation of miniaturised hydrogen fuel cells.

Compared to Li-ion or LiPo batteries, hydrogen fuel cells offer many advantages, primarily their high energy to weight ratio. The engineers at DMI utilised very thin metals with a proprietary technology for the bipolar plates in the fuel cell. Known as DP30, the fuel cell has two main powertrains that